EDS THEORY ASSIGNMENT: 2

NAME: Gunjan Swami

PRN: 202501120128

1. Identifying Data Completeness
Problem: Check for null or missing values in all columns.
Solution: df.isnull().sum()

2. Verifying Unique Blog Posts
Problem: Count unique blog entries.
Solution: unique_count = df['text'].nunique()

3. Gender Distribution
Problem: Count each gender category.
Solution: gender_counts = df['gender'].value_counts()

4. Age Group Density
Problem: Calculate percentage of posts by a specific age group.
Solution: age_percent=(df['age']==17).mean()*100

5. Measuring Blog Length
Problem: Add total character length column.
Solution: df['char_length']=df['text'].apply(len)

6. Vocabulary Size Analysis
Problem: Count words in each blog.
Solution: df['word_count']=df['text'].apply(lambda x:len(x.split()))

7. Comparative Length Analysis
Problem: Compare average blog length by gender.
Solution: avg=df.groupby('gender')['char_length'].mean()

8. Detecting Extremes
Problem: Find maximum blog length.
Solution: max_len=df['char_length'].max()

9. Statistical Spread of Words
Problem: Standard deviation of word count.
Solution: word_std=np.std(df['word_count'])

10. Binary Encoding for Models
Problem: Encode gender numerically.
Solution: df['is_male']=np.where(df['gender']=='male',1,0)

11. Filtering Emotional Content
Problem: Extract blogs containing 'love'.
Solution: love=df[df['text'].str.contains('love',case=False)]

12. Word-to-Character Ratio
Problem: Avg chars per word.
Solution: df['avg_word_len']=df['char_length']/df['word_count']

13. Identifying Outliers
Problem: Find blogs beyond 99th percentile length.
Solution: t=np.percentile(df['char_length'],99); out=df[df['char_length']>t]

14. Correlation Between Length and Gender
Problem: Correlation of gender encoding and length.
Solution: corr=df['is_male'].corr(df['char_length'])

15. Data Deduplication
Problem: Remove duplicate blogs.
Solution: df=df.drop_duplicates(subset='text')

16. Numeric Content Detection
Problem: Count blogs containing digits.
Solution: digit=df['text'].str.contains(r'\d').sum()

17. Most Common Blog Lengths
Problem: Top 3 frequent lengths.
Solution: top=df['char_length'].value_counts().head(3)

18. Short Blog Filtering
Problem: Count blogs with <=10 words.
Solution: short=df[df['word_count']<=10].shape[0]

19. Aggregated Statistics
Problem: Min, max, median length by gender.
Solution: stats=df.groupby('gender')['char_length'].agg(['min','max','median'])

20. Sampling for Inspection
Problem: Select 5 random blogs.
Solution: sample=df.sample(5)